
Problem 10.21. Let $EQ_{BP} = \{\langle B_1, B_2\rangle \mid B_1 \text{ and } B_2 \text{ are equivalent branching programs}\}$. Show that EQ_{BP} is coNP-complete.

Proof. To show that EQ_{BP} is coNP-complete, we must show that it is in coNP and that all coNP-problems are polynomial time reducible to it. $\overline{EQ_{BP}}$ is in NP as the certificate is simply an assignment of truth values to variables on which two branching programs give different outputs. Therefore, EQ_{BP} is in coNP. To prove the second part, we show that $\overline{SAT}$ is polynomial time reducible to EQ_{BP}. The reduction converts a Boolean formula ϕ into two branching programs B_1 and B_2, so that ϕ is not satisfiable, iff B_1 and B_2 are equivalent. B_1 is constructed to implement ϕ's Boolean function. Such construction can be done in polynomial time. B_2 is constructed to always output 0. □